and molar mass of ethylene glycol is 62 g mol^{-1}.

Weight of solute (W_2) = 20 mass percent of solution means 20 g of ethylene glycol

Weight of solvent (water) W_1 = 100 -20 = 80 g

$\Delta T_f = K_f m$

$$= \frac{K_f \times W_2 \times 1000}{M_2 \times W_1}$$

$$= \frac{1.86 \times 20 \times 1000}{62 \times 80}$$

= 7.5 K

The temperature at which ice will begin to separate is the freezing point of water after the addition of solute i.e 7.5 K lower than the normal freezing point of water (273-7.5K) = 265.5 K

Evaluate Yourself

12) 2g of a non electrolyte solute dissolved in 75 g of benzene lowered the freezing point of benzene by 0.20 K. The freezing point depression constant of benzene is 5.12 K Kg mol^{-1}. Find the molar mass of the solute.

Osmosis and osmotic pressure

Many biological processes depend on osmosis, which is a spontaneous process by which the solvent molecules pass through a semi permeable membrane from a solution of lower concentration to a solution of higher concentration. The name osmosis is derived from the Greek word '*osmos*' which means 'to push'. It is also important to know that the semipermeable membrane selectively allows certain molecules in the solution to pass through it but not others.

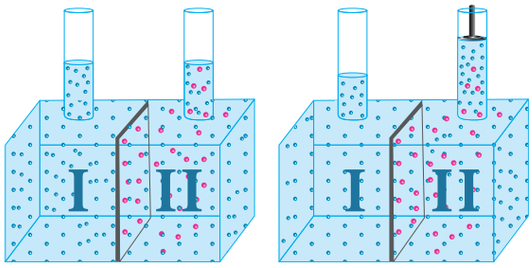


Figure 9.12 Osmosis and osmotic pressure

Let us consider a simple apparatus as shown in the above figure. A semipermeable membrane separates a chamber into two compartments. Water (pure solvent) is added to the first compartment and the aqueous NaCl (solution) is added to the second compartment such that the liquid levels on the both sides are equal. Since there is a difference in concentration between the liquids present in the two compartments, the water molecules move from first compartment to second compartment through the semipermeable membrane. The membrane allows only water molecules to pass through it in either direction but not NaCl. The net flow of water is into the sodium chloride solution and hence increases its volume. This decreases its concentration and also creates a pressure difference between the compartments. This pressure difference, push some of the water molecules back to the solvent side through the semipermeable membrane until an equilibrium is established. At the equilibrium, the rate of movement of solvent molecules on both directions are equal. The pressure difference at the equilibrium is called osmotic pressure (π). Thus, osmotic pressure can be defined as "the pressure

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